

User Performance in Consecutive Temporal Pointing: An Exploratory Study

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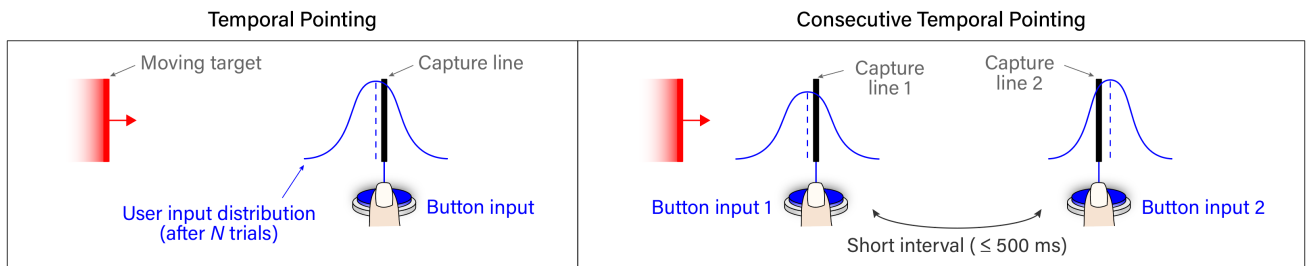


Figure 1: (Left) Temporal pointing requires users to generate a single button input at a designated timing. **(Right) Consecutive temporal pointing (CTP)** requires users to generate two consecutive button inputs while satisfying temporal constraints for each. Little is known about the user performance in CTP tasks.

ABSTRACT

A significant amount of research has recently been conducted on user performance in so-called *temporal pointing* tasks, in which a user is required to perform a button input at the timing required by the system. Consecutive temporal pointing (CTP), in which two consecutive button inputs must be performed while satisfying temporal constraints, is common in modern interactions, yet little is understood about user performance on the task. Through a user study involving 100 participants, we broadly explore user performance in a variety of CTP scenarios. The key finding is that CTP is a unique task that cannot be considered as two ordinary temporal pointing processes. Significant effects of button input method, motor limitations, and different hand use were also observed.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

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KEYWORDS

Temporal Pointing, Fitts' Law, Button Inputs

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1 INTRODUCTION

A button, in a general sense, refers to a transducer that registers the motion of a body part (e.g., a finger), changes the state of a computer, and returns to a resting state [30]. Under this definition, a button does not mean only the physical button we are familiar with, but encompasses all implementations that can deliver discrete events to the computer, such as touch [20] and mid-air gestures [10, 22]. As everyone experiences, activating a button usually does not require much effort. However, in modern human-computer interaction (HCI), making button inputs can sometimes be a very challenging and difficult task. For example, suppose a moving target appears on the screen and users have to press a button when it touches a capture line (see Figure 1, Left). The speed of the target is fast enough so that the time the target and the line are in contact is very short (i.e., less than 200 ms). Then users should not perform button input after confirming the contact of the target and the line, because the target has already left the line after the reaction time (typically longer than 250 ms) [33] has elapsed. Instead, they

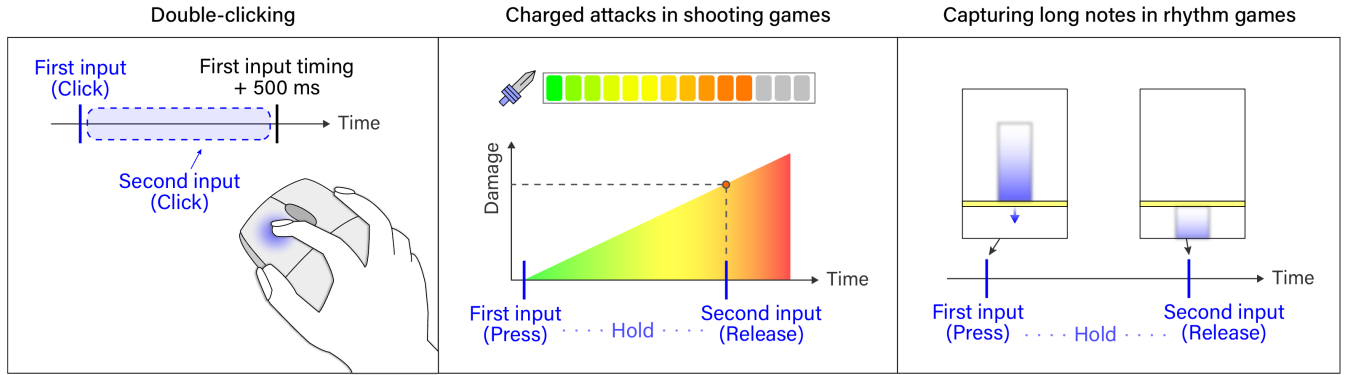


Figure 2: Examples of CTP tasks - For double-clicking and charged attacks, the second button input should be performed at the exact moment after the first button input. For long notes in rhythm games, each of the two button inputs should be performed at a specific timing.

have to *anticipate* when the target will hit the line in advance [37, 40, 41], and start the button input execution before the target actually touches the line.

As in the previous example, the task of anticipating and synchronizing a button input at a timing specified by the system is called *temporal pointing*, and its user performance has been actively studied since 2016 in HCI [5, 9, 16–21, 23, 24, 26, 39, 42–45]. In temporal pointing tasks, the distribution of button input timing of a user generally shows a Gaussian distribution around the timing specified by the system [23, 24, 26]. The smaller the standard deviation σ of the button input distribution and the closer the mean μ is to the timing specified by the system, the higher the user performance we can consider. Several studies [23, 24, 26, 30, 31] published in the past few years have identified cognitive mechanisms in which humans perform temporal pointing, and some studies have presented scientific models that predict user performance with a high coefficient of determination (R^2 ranging from 0.81 to 0.99) [21, 23, 24, 26].

Meanwhile, users often also need to generate two consecutive anticipatory button inputs with specific timing requirements for each input and a short time interval (typically less than 500 ms) between the two inputs, considering them as a *single chunk*. We define such tasks as *consecutive temporal pointing* (CTP). CTP tasks are easily found in video games. To execute the charged attack in shooting games, users typically need to press (the first input) and hold a button to charge it up, and then release (the second input) the button to unleash it. Users should generate the two button inputs at specific timings to produce the attack at a particular moment with their intended amount of damage. This also applies to capturing a long note (or held note) in rhythm games. To maintain the combo, the two button inputs should be generated exactly at each timing when the note hits the target bar and when it leaves the bar, respectively (see Figure 2). Even outside of the game context, several interaction techniques based on CTP are widely used or have been proposed: double-clicking [14, 15, 29], Rhythmic Menus [27], and single-switch based target selection [3, 4, 7].

While CTP is extensively employed in many HCI scenarios, there is currently no scientific model to predict user performance in CTP,

even the empirical understanding of user performance in this context has barely been touched upon. The *key question* is whether CTP can be considered just two separate, ordinary temporal pointing tasks. If so, the existing theories and models of temporal pointing [21, 23, 24, 26] can be applied as they are. To fill this gap, in this study we provide the first holistic empirical understanding of user performance in CTP. We observed performance from a total of 100 participants in a user study that encompassed a wide range of known factors of temporal pointing, such as input period [23, 24, 26], cue-viewing time [23, 24, 26, 31], and button activation point [20, 24].

As a result, we confirmed that CTP is a *unique task* that cannot be considered as two separate ordinary temporal pointing processes; we found the existing state-of-the-art temporal pointing model [23, 24, 26] failed to predict user performance in CTP, especially for the second button input. Furthermore, we found interesting phenomena such as the second button input lags as the input interval becomes shorter due to motor limitations, and the `keyPressed` callback function gives higher temporal precision than `keyReleased`. We expect these new findings to spark future modeling research, and to maximize impact, we are releasing all datasets, experiment code, and analysis code as open source¹. We conclude by discussing the implications of our findings for game design and interaction design.

2 CONSECUTIVE TEMPORAL POINTING: AN OVERVIEW

In this section, we present a formal definition of what a CTP task is, along with a general taxonomy, and explain how user performance on the task can be generally quantified.

2.1 Definition and Taxonomy

2.1.1 Definition. We define CTP as a task that satisfies the following conditions: (1) users must perform two consecutive button inputs separated by a short time interval, typically less than 500

¹<https://w-dlee.github.io/ctp>

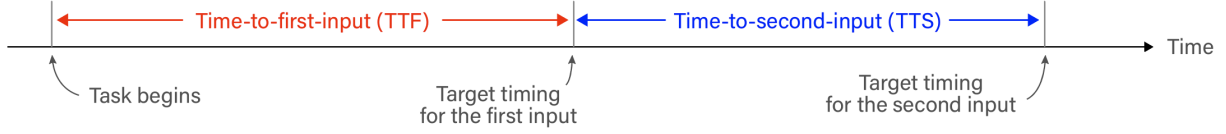


Figure 3: Task variables in CTP: Time-to-First-Input (TTF) and Time-to-Second-Input (TTS)

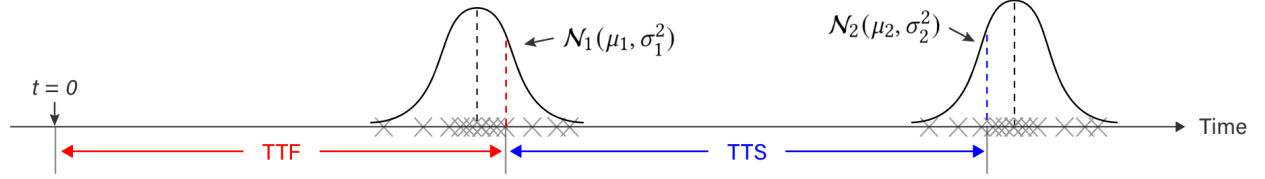


Figure 4: The Gaussian distributions of the first and second button inputs in CTP: user performance is evaluated by the difference in the average of their button inputs (μ) with the target time for the input, as well as the spread of the user inputs (σ).

ms², (2) the system requests that the timing of the two button inputs satisfy certain temporal conditions (i.e., when the button input should be generated), and (3) users successfully complete the task when they satisfy all the temporal conditions required for both button inputs.

Figure 3 shows the timing of two consecutive button inputs that the system requests from a user on the time axis. Let us call the time from the start of the task to the occurrence of the first button input *time-to-first-input* (TTF), and the time from the first button input to the second button input *time-to-second-input* (TTS).

In temporal pointing, the final goal of users is to perform button input *as close as possible* to the timing requested by the system. According to previous studies [20, 23], the distribution of user button input timing in temporal pointing tasks generally follows a Gaussian. From the mean (μ , accuracy) and standard deviation (σ , precision) of the distribution, user performance in the temporal pointing task can be quantified. There are two button inputs in CTP tasks, so input accuracy and precision must be measured independently for each button input. In this study, the Gaussian distributions observed from the first and second button inputs are denoted as $N_1(\mu_1, \sigma_1^2)$ and $N_2(\mu_2, \sigma_2^2)$, respectively (see Figure 4).

2.1.2 Taxonomy. According to the patterns in which temporal requirements from the system are given, CTP tasks can be classified into two types (see Figure 5). The *Type I CTP* refers to cases in which the system imposes temporal requirements only on the second button input. In the task, users can perform the first button input at any time, and the trial of the task starts from that moment. Instead, the second button input must be performed while satisfying a specific temporal condition with respect to the moment the first button input. The temporal condition imposed may be relatively loose or, conversely, very strict. For example, in the process of

executing an application by double-clicking an icon, it is sufficient if the second button input is simply performed before a certain time elapses from the first input. On the other hand, in examples found in popular video games, such as determining the speed at which a ball is launched in Pinball or determining how much a character jumps in Jump King, the second button input needs to be performed in synchronization as close as possible to the specific moment that comes after the first input.

The *Type II CTP* refers to cases where the system imposes temporal requirements on both button inputs. A trial is triggered by an external event, and users must perform the first button input and the second button input accurately in time with respect to the starting moment of the trial. The most typical example is the long note given to players in rhythm games (see Figure 2). To successfully capture the long note, players must make button inputs correctly both at each moment when the bottom and top of the long note, which is usually moving down, touches the judging line.

We can also imagine a case in which the temporal requirement is imposed only on the first button input, not on the second. However, in that case, users only need to focus on the success of the first button input, which brings it closer to the ordinary temporal pointing task performed with a single button input. On the other hand, if no temporal requirements are imposed on both button inputs, the task is no longer temporal pointing and can be considered a normal consecutive button input task. Figure 5 summarizes the general taxonomy of consecutive button input tasks.

3 BACKGROUND & RESEARCH QUESTIONS

In this section, we present a summary of previous studies and theories related to temporal pointing. Then we explain what insights those studies provide on user performance in CTP tasks, and derive key research questions for this study.

²If the time interval between two button inputs is long enough, each button input can be performed independently of the other and consequently cannot be considered a case of consecutive button input tasks.

		Time-to-first-input (TTF)	
		Temporally unconstrained	Temporally constrained
Time-to-second-input (TTS)	Temporally unconstrained	Ordinary consecutive button input task	Ordinary temporal pointing task
	Temporally constrained	Consecutive temporal pointing task (<i>Type I</i>)	Consecutive temporal pointing task (<i>Type II</i>)

Figure 5: Taxonomy of consecutive button input tasks

3.1 Introduction to Temporal Pointing

In the process of interacting with a computer, users generally perform numerous target acquisition tasks. Representative examples include clicking or touching an icon on a screen or a touchscreen, navigating an upper menu of a window with a pointer and selecting a target item. In all these examples, the target that users want to acquire is implemented *in space*. In other words, to acquire a target, the pointer (or finger in the case of a touchscreen) must be moved to a specific location in space. HCI researchers usually refer to this kind of target acquisition task as *pointing*. Research on user performance in the pointing task has been actively conducted since the early days of HCI, and robust and accurate user performance prediction models such as Fitts' law [12] are widely known.

Temporal pointing is a concept proposed by Lee and Oulasvirta [24] in 2016 and deals with target acquisition *in time* rather than space. In temporal pointing, users are required to perform button input at a specific location in time (i.e., timing) rather than a specific location in space (see Figure 6). Temporal pointing is ubiquitous in applications such as music and real-time video games, and along with spatial pointing (explained by Fitts' law), constitutes the most fundamental input task of HCI.

Two representative ways of implementing a temporal pointing task are known: (1) making the target repeatedly appear and disappear in a user-predictable way (e.g., a target blinking at a specific interval) and (2) making the target move in a predictable way towards a specific acquisition point in space (e.g., a note moving to the bottom judgment line in rhythm games). More generally, it is also possible to implement a mixture of these two methods (e.g., repeatedly creating a moving target).

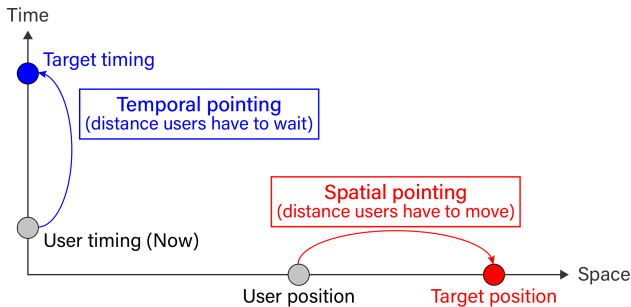


Figure 6: Conceptual diagram of spatial pointing and temporal pointing tasks

3.2 Factors Affecting User Performance in Temporal Pointing

Although it seems like a simple task to press a button at a specific timing, user performance in temporal pointing varies widely due to several factors. In this section, we present a summary of the major factors affecting user performance in temporal pointing.

3.2.1 Input Period. If the target appears repeatedly with a specific time period P , users can guess when the next button input should be performed (Figure 7a). In this process, the precision of the estimation, that is, the reciprocal of the standard deviation σ of the button input distribution, drops in proportion to the length of P^3 :

$$\sigma = c_1 P. \quad (1)$$

Cognitive scientists explain that an internal clock, a kind of stopwatch, is involved in the process by which humans encode the time intervals between repeated events. The above equation represents the unique noise characteristics of humans in time interval encoding and is generally called the *scalar property* of the internal clock [13, 40, 41]. c_1 is a parameter representing the difference in the precision of the internal clock between individuals.

3.2.2 Cue-Viewing Time. Suppose the target is moving in a predictable pattern. Users can then estimate the timing at which the target will arrive at a specific location in space (where target acquisition must be made). In this process, the key variable that determines the precision of button input is the given duration to observe the target movement (i.e., cue-viewing time t_c) [23, 26, 31]. The longer t_c , the better the user encodes the movement pattern of the target and more precise button input can be performed (i.e., lower σ , see Figure 7b). This effect can be approximated by the equation:

$$\sigma = c_2 + \frac{1}{e^{c_3 t_c} - 1}. \quad (2)$$

Here c_2 represents the amount of floor noise that still remains even when t_c is long enough. The c_3 represents the rate at which precision converges when t_c increases. The c_2 and c_3 are parameters representing the difference in encoding performance between individuals or the encoding difficulty of a specific target motion.

3.2.3 Button Activation Point. In the button input process, a user's continuous input movement is converted into a discrete event and transmitted to the computer. In this process, the reference point at which the user's movement is converted into a discrete event is called the *activation point* of the button [20]. For example, typical physical keyboard buttons can be set to be activated either while the button is being pressed (i.e., `keyPressed` callback function) or while it is being released (i.e., `keyReleased` callback function).

According to previous studies [20, 24, 30], button input precision can be improved when the activation point of a button is aligned with the point where users feel the greatest sensory stimulation during input movement. In the case of a physical button, a strong impact is transmitted to the finger the moment the button cap hits the floor, so when using the `keyPressed` callback function rather than the `keyReleased`, users' button input precision can be greatly improved (upto 7.5%p) [24] (Figure 7c). A similar effect can be

³Recall that clapping once every five seconds is much harder to be precise than clapping once every one second.

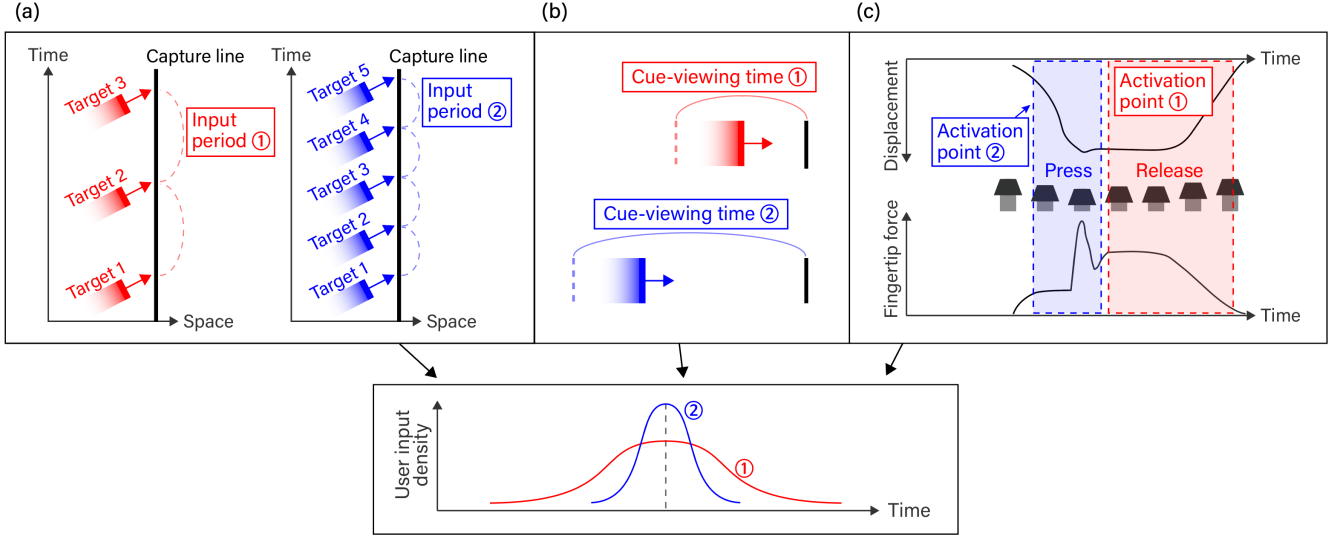


Figure 7: Three factors generally determining user performance in temporal pointing: (a) repetition period of the input, (b) cue-viewing time, and (c) button activation point

observed for touch buttons [24], and a theoretical explanation for this effect was presented in a paper published in 2018 [30].

3.3 Predicting Temporal Pointing Performance

Since the concept was first proposed in 2016, there have been several studies that have presented mathematical models that predict user performance in temporal pointing or related tasks [9, 16–19, 39, 42–45]. A model that can be applied to the most general type of temporal pointing task, presents a clear cognitive mechanism, and can accurately predict user performance was presented by Lee et al. [23] in 2018. The model deals with a general temporal pointing scenario called *moving-target acquisition* (MTA), in which a target that is repeatedly generated at a constant time period moves toward a specific acquisition point (Figure 8). The model can predict the standard deviation σ of the button input distribution in the task with a high coefficient of determination ($R^2 = 0.812$) as a function of the following two variables: (1) target creation period P and (2) time required for the created target to move to the acquisition point (i.e., cue-viewing time t_c). The exact equation of the model is:

$$\sigma = \frac{c_1 P (c_2 + \frac{1}{e^{c_3 t_c} - 1})}{\sqrt{(c_1 P)^2 + (c_2 + \frac{1}{e^{c_3 t_c} - 1})^2}}. \quad (3)$$

Here, c_1 , c_2 , and c_3 are free parameters of the model and represent user-specific characteristics. To the best of our knowledge, this model can be regarded as a baseline model of user performance in temporal pointing.

Meanwhile, a model capable of predicting the mean μ of the user button input distribution in temporal pointing does not yet exist; only heuristic rules have been proposed [23, 24, 26].

3.4 Research Questions

Reflecting on previous studies of temporal pointing, our research questions (RQs) in this study were derived as follows:

- **RQ1.** In a CTP task, can the first and second button inputs be considered an ordinary temporal pointing process?
- **RQ2.** How do factors known to significantly affect user performance in ordinary temporal pointing (e.g., button activation point) affect user performance in CTP?

4 USER STUDY: THREE VARIATIONS OF CTP

To answer the research questions, we conducted a user study with a total of 100 participants. The user study covers three different variations of the CTP task. In the first variation, temporal constraints were given only to the second button input (*Type I CTP*). In the second variation, temporal constraints were given to both button inputs (*Type II CTP*). In the third variation, both *Type I* and *II* tasks were performed, but buttons were pressed with both hands.

4.1 Task

The task performed by participants in all three variations was identical: *moving-target acquisition* [23, 26], the most generalized form of temporal pointing. In the task, participants are given a line that appears from the left and moves with constant velocity to the right. Participants must make synchronized button inputs at the moment the line touches each capture line (Figure 9). The moving line is created repeatedly at the same location at a specific period.

4.2 Design

Task design is determined from six independent variables: (1) TTF (Time-to-First-Input), (2) TTS (Time-to-Second-Input), (3) Input Period, (4) Visual Cue Presence, (5) Input Method, and (6) Input

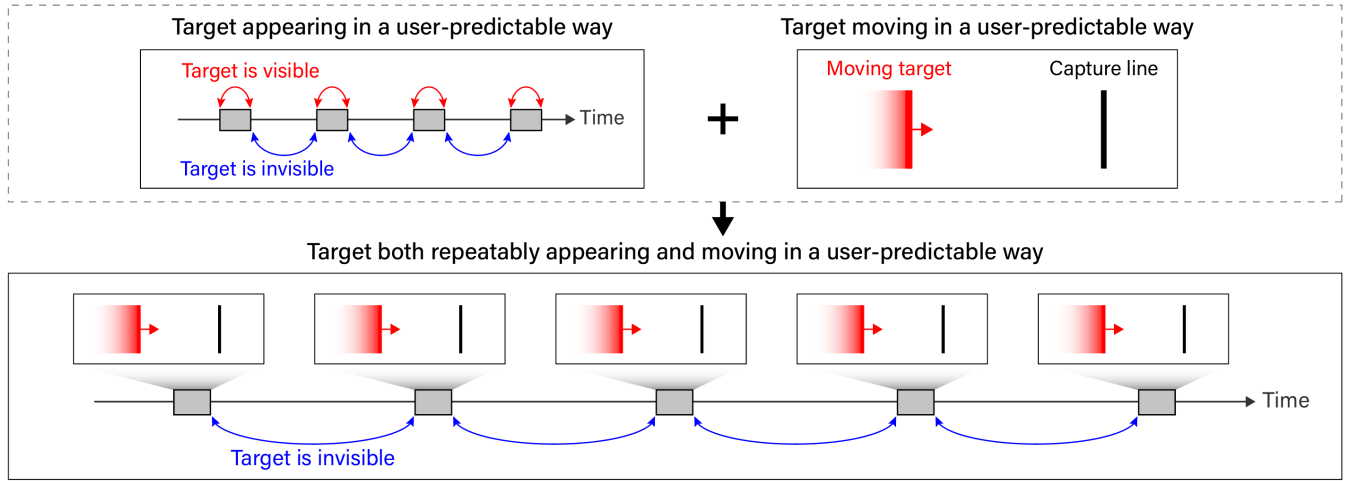


Figure 8: Conceptual diagram of the most general temporal pointing task, so-called moving-target acquisition, where the target both repeatably appears and moves in a user-predictable way

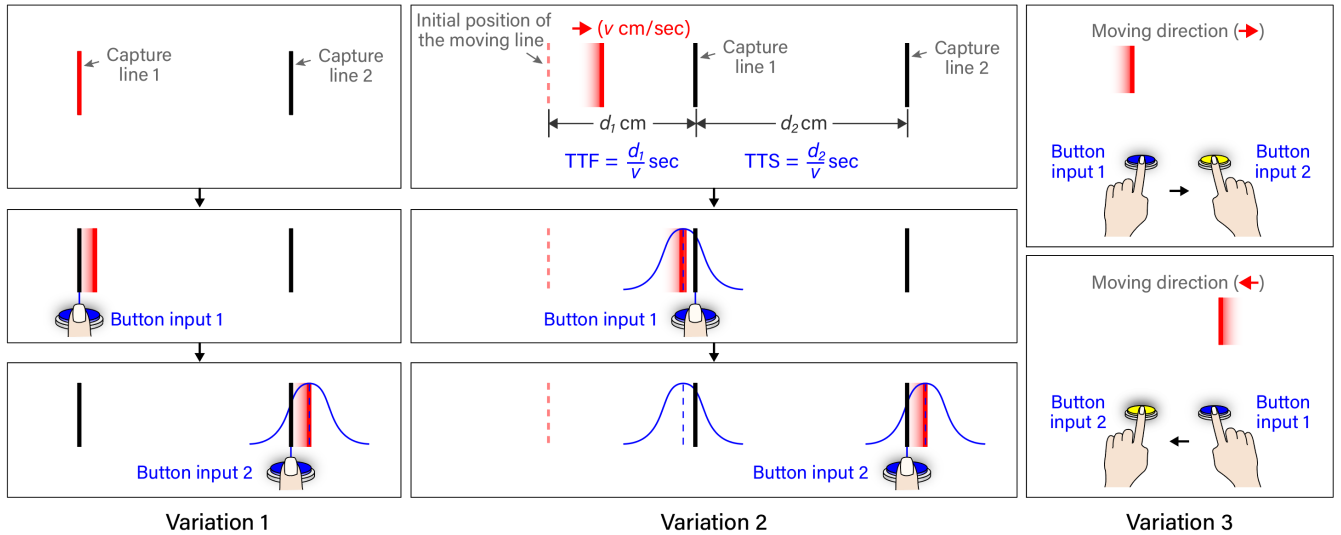


Figure 9: Task overview of the three variations in the user study - In Variation 1, the user can initiate the movement of the target (red line) by performing the first button input. In Variation 2, the target appears periodically. In Variation 3, the tasks of Variations 1 and 2 are performed using two buttons bimanually.

Order. Depending on the task variation, only some of these independent variables may be included.

TTF is the time from when the moving line appears until it reaches the first capture line. TTS is the time from the moment the moving line touches the first capture line until it touches the second capture line. TTF and TTS can be adjusted from the positions of the capture lines when the speed of the moving line is fixed (28.2 cm/sec in this study). Please refer to Figure 9 for details on how TTF and TTS are calculated. Input Period indicates the period in which the moving line is created repeatedly. TTF and

Input Period are not definable in *Type I CTP* due to the nature of the task, where participants voluntarily make the first button input.

Visual Cue Presence is a variable that determines whether the line is visible in the process of reaching the capture line and has two levels: On or Off (see Figure 10). Input Method is a variable that determines which activation point each of the two button inputs has and has two levels: Press-Press or Press-Release. In the Press-Press condition, both button inputs are activated through the keyPressed callback function. In the Press-Release condition, the second button

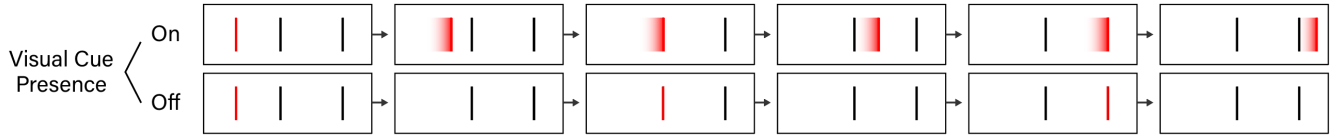


Figure 10: Two levels of Visual Cue Presence: In On condition, the user can see the real-time movement of the moving line. In Off condition, the moving line only appears briefly (for 20 ms) when it is first created and when it contacts each capture line.

Table 1: Experimental design and demographic information of participants in three variations of the user study

	Variation 1	Variation 2	Variation 3.1	Variation 3.2
Type	<i>Type I CTP</i>	<i>Type II CTP</i>	<i>Type I CTP</i>	<i>Type II CTP</i>
Design	8 TTS × 2 Visual Cue Presence × 2 Input Method	2 TTF × 3 TTS × 2 Input Period × 2 Visual Cue Presence* × 2 Input Method	8 TTS × 2 Visual Cue Presence × 2 Input Order	2 TTF × 3 TTS × 2 Visual Cue Presence × 2 Input Order
TTF	—	150 or 250 ms	—	150 or 250 ms
TTS	50, 100, 150, 200, 250, 300, 400, or 500 ms	150, 200, or 250 ms	50, 100, 150, 200, 250, 300, 400, or 500 ms	150, 200, or 250 ms
Input Period	—	1000 or 1500 ms	—	1000 ms
Visual Cue Presence	On or Off	On or Off	On or Off	On or Off
Input Method	Press-Press or Press-Release	Press-Press or Press-Release	Press-Press	Press-Press
Input Order	—	—	Dominant-to-NonDominant or NonDominant-to-Dominant	Dominant-to-NonDominant or NonDominant-to-Dominant
The Number of Participants	25 (8 females)	On: 23 (11 females) Off: 20 (12 females)	16 (6 females)	16 (3 females)
Mean Age of Participants	26.32 (SD = 3.86)	25.98 (SD=3.62)	22.50 (SD=2.53)	22.25 (SD=2.70)
Handedness	Two left-handed	Three left-handed	All right-handed	Two left-handed
Total Data	32,000 (25 participants × 40 trials × 32 conditions)	41,250 (43 participants × 40 trials × 24 conditions)	20,480 (16 participants × 40 trials × 32 conditions)	15,360 (16 participants × 40 trials × 24 conditions)

* indicates a between-subject factor. Factors without this mark are within-subject factors.

input is changed to use the keyReleased callback function. Lastly, Input Order refers to the order in which two button inputs are performed alternately with both hands and has the following two levels: Dominant-to-NonDominant or NonDominant-to-Dominant. The direction of line movement is set to be compatible with the order of the hands.

As a result, three different variations of the user study were conducted, each with different participants and a different design. Table 1 summarizes demographic information of participants and design of each variation. All independent variables were within-subjects factors, except for Visual Cue Presence in Variation 2, which was set to between-subjects in order to keep the experimental time within reasonable limits. We recruited at least 16 participants for each variation, which falls within the range of typical CHI studies [8] and the temporal pointing literature [21, 23, 24, 26].

All button input signals were logged in all variations. Participant performance was quantified as the mean (μ_1 and μ_2) and standard

deviation (σ_1 and σ_2) of the distribution of each button input (see Figure 4). All participants were recruited from local universities.

4.3 Procedure and Apparatus

The experiment took an average of one hour, and the conditions were counterbalanced. To minimize the input latency, high performance computers and a custom-designed button were used (Figure 11). Participants were asked to perform 50 trials per condition, but considering their learning, the last 40 trials were used for the analysis. If the timing error was too large, where the absolute difference between TTS and the logged time interval of the two button inputs exceeds 120 ms, we requested participants to perform that trial again. For other details on the experimental procedure and apparatus, please refer to Supplementary Material A.

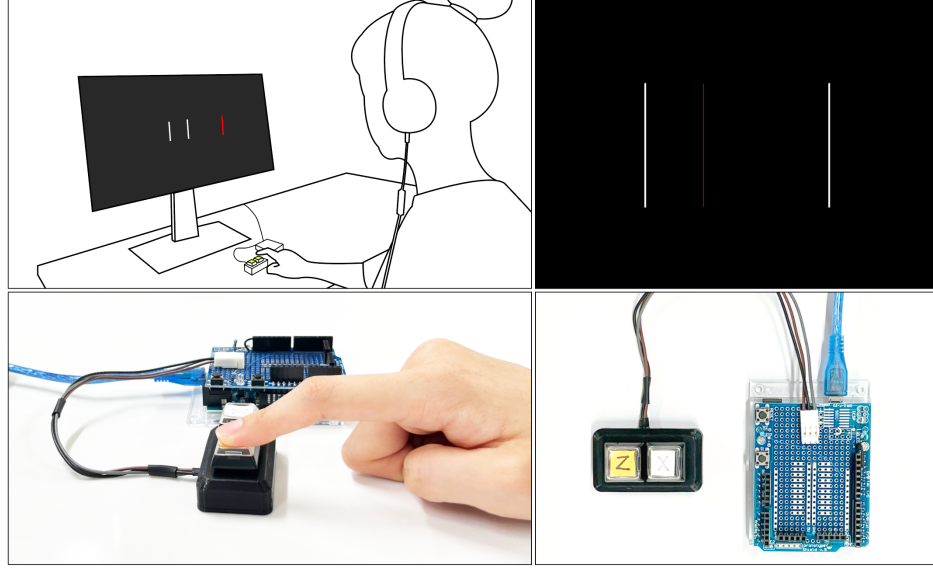


Figure 11: The experimental setting (upper left), the task screen (upper right), and the low-latency button (lower left, right)

5 RESULTS

In this section, we report general descriptive statistics, model fitting attempts to check whether the existing temporal pointing models are applicable in CTP, and the statistical significance test results of the effect of independent variables on button input distributions.

5.1 Descriptive Statistics

A total of 109,090 trials were collected. Among these trials, 2,649 ($\approx 2.43\%$) were recognized as unacceptable input errors and requested to be performed again. For additional descriptive statistics and the results of the normality test, refer to Supplementary Material B.

5.2 Model Fitting

If CTP is a succession of two ordinary temporal pointing processes, a model of temporal pointing should be able to successfully predict the standard deviation of the first and second button inputs in CTP. We performed model fitting on the average values of σ_1 and σ_2 observed from each participant in the study to the state-of-the-art temporal pointing model (Equation 3) [21, 23]. Details of the model fitting process are in Supplementary Material C.

The temporal pointing model fitting was successful for σ_1 in Variation 2 (see Figure 12a, Left) with $R^2 = 0.93$; $c_1 = 0.08$, $c_2 = 0.03$, and $c_3 = 29.72$. Due to having only two data points in Variation 3.2, which is fewer than the number of parameters in the model, fitting was impossible (four data points were used in Variation 2). However, we still found a similar trend to the temporal pointing model, where σ decreases as t_c (i.e., TTF) increases (see Figure 12a, Right). However, the temporal pointing model could not be fitted to σ_2 (see Figure 12b). σ_2 exhibited different trends, where σ does not decrease as t_c (i.e., TTS) increases, making the fitting impossible.

5.3 Statistical Analysis

For the statistical analysis, repeated measures ANOVA was performed using JASP version 0.16.4, with a significance level of $\alpha = 0.05$. If the sphericity assumption was violated, a Greenhouse-Geiser correction was applied. Bonferroni correction was applied to all post-hoc analyses. Statistically significant effects are summarized in Tables 2, 3, 4, and 5. A full table including effects that were not statistically significant is in Supplementary Material D.

6 DISCUSSION

In this section, we answer the RQs posed in Section 3.4 based on the experimental results. We further discuss general implications of this study in interaction design, limitations, and future directions for follow-up research.

6.1 Can CTP Be Considered Two Ordinary Temporal Pointing Processes? (RQ1)

Based on the results, it is hard to regard CTP as a combination of two ordinary temporal pointing processes. According to the temporal pointing model [21, 23, 26], σ should decrease (getting more precise) as t_c increases (i.e., Equation 2), as in the case of σ_1 (see Figure 12a). However, contrary to the model, σ_2 actually increased as t_c (i.e., TTS) became longer (see Figure 12b). There appear to be hidden cognitive factors in the second button input process that the existing model cannot explain.

One possible explanation for what makes CTP such a unique task is that participants utilized the time interval between two button inputs, TTS, as an additional sensory cue to determine the timing of the second button input. In other words, just as the timing of the next button input can be estimated from the overall repetition period P of the trials, by encoding TTS with the internal clock, participants can determine the timing of the second button input

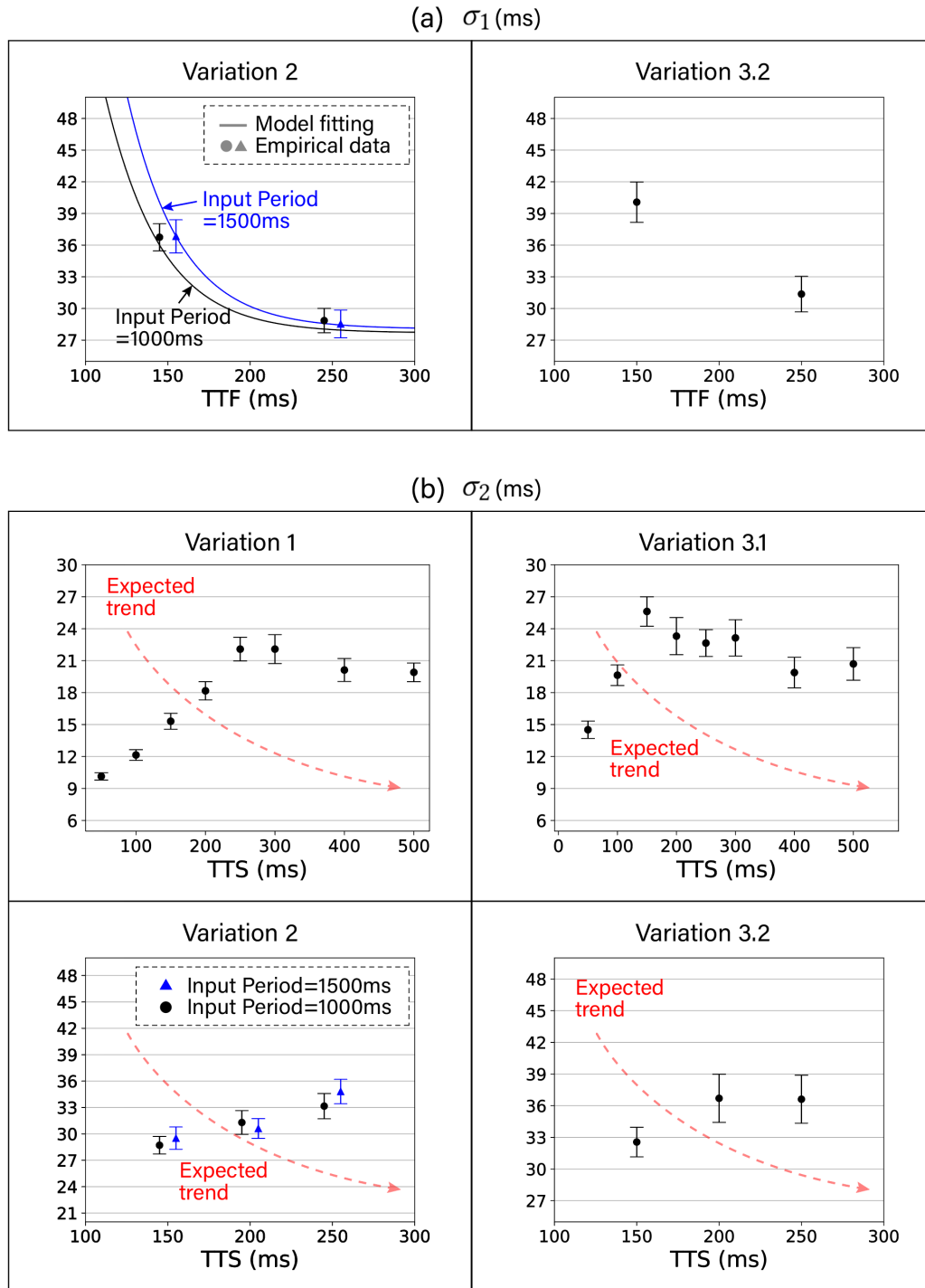


Figure 12: Results of fitting CTP experimental data to the temporal pointing model in each study variation - σ_1 exhibited a decreasing trend that was expected from the temporal pointing model, while σ_2 did not conform to the expected trend. The error bars represent the standard error of the mean.

Table 2: Significant main effects found in Study Variation 1**Study Variation 1**

	Factor	df	df_{res}	F	p	η_p^2
μ_2	TTS	1.7	40.3	143.6	<.001***	0.857
	TTS $\times$ Visual Cue Presence	3.7	88.5	6.8	<.001***	0.222
	TTS $\times$ Input Method	2.6	63.0	120.4	<.001***	0.834
	Input Method	1.0	24.0	105.8	<.001***	0.815
σ_2	TTS	3.3	78.7	126.0	<.001***	0.840
	TTS $\times$ Visual Cue Presence	4.9	116.8	37.8	<.001***	0.612
	Visual Cue Presence	1.0	24.0	73.6	<.001***	0.754
	Visual Cue Presence $\times$ Input Method	1.0	24.0	5.2	0.032*	0.178
	Input Method	1.0	24.0	33.0	<.001***	0.579

*, **, and *** indicate $p < .05$, $p < .01$, and $p < .001$, respectively.

Table 3: Significant main effects found in Study Variation 2**Study Variation 2**

	Factor	df	df_{res}	F	p	η_p^2
μ_1	TTF	1.0	-	265.8	<.001***	0.866
	TTF $\times$ Input Period	1.0	-	30.0	<.001***	0.423
	TTS	1.6	-	6.9	0.004**	0.144
	TTS $\times$ Input Method	2.0	-	5.6	0.005**	0.120
	Input Period	1.0	-	55.0	<.001***	0.573
σ_1	TTF	1.0	-	151.9	<.001***	0.787
	TTF $\times$ TTS	1.8	-	4.2	0.022*	0.093
	TTF $\times$ Input Period	1.0	-	13.5	<.001***	0.247
	TTF $\times$ Input Method	1.0	-	5.3	0.027*	0.114
	Input Period	1.0	-	16.2	<.001***	0.283
	Input Period $\times$ Visual Cue Presence	1.0	41.0	14.1	<.001***	0.255
	Visual Cue Presence	1.0	41.0	9.2	0.004**	0.183
μ_2	TTF	1.0	-	6.9	0.012*	0.145
	TTF $\times$ TTS	1.9	-	7.4	0.001***	0.153
	TTF $\times$ Input Period	1.0	-	7.3	0.010*	0.152
	TTF $\times$ Input Method	1.0	-	6.0	0.019*	0.127
	TTS	1.7	-	35.7	<.001***	0.466
	TTS $\times$ Input Method	1.4	-	9.6	0.001**	0.190
	Input Period	1.0	-	16.5	<.001***	0.287
	Input Period $\times$ Input Method	1.0	-	8.3	0.006**	0.168
σ_2	TTF	1.0	-	5.3	0.026*	0.115
	TTF $\times$ TTS	1.9	-	3.7	0.031*	0.082
	TTF $\times$ Visual Cue Presence	1.0	41.0	20.5	<.001***	0.334
	TTF $\times$ Input Method	1.0	-	11.7	0.001**	0.222
	TTS	2.0	-	19.2	<.001***	0.319
	Input Period	1.0	-	9.1	0.004**	0.181
	Visual Cue Presence	1.0	41.0	7.8	0.008**	0.160
	Input Method	1.0	-	12.6	0.001**	0.234

*, **, and *** indicate $p < .05$, $p < .01$, and $p < .001$, respectively.

Table 4: Significant main effects found in Study Variation 3.1**Study Variation 3.1**

	Factor	df	df_{res}	F	p	η_p^2
μ_2	TTS	4.1	61.1	23.3	<.001***	0.608
	TTS $\times$ Visual Cue Presence	3.6	53.5	4.1	0.007**	0.216
	Visual Cue Presence	1.0	15.0	5.8	0.029*	0.280
σ_2	TTS	4.1	61.6	37.6	<.001***	0.715
	TTS $\times$ Visual Cue Presence	3.6	54.4	15.0	<.001***	0.499
	Visual Cue Presence	1.0	15.0	39.2	<.001***	0.723

*, **, and *** indicate $p < .05$, $p < .01$, and $p < .001$, respectively.

Table 5: Significant main effects found in Study Variation 3.2**Study Variation 3.2**

	Factor	df	df_{res}	F	p	η_p^2
μ_1	TTF	1.0	15.0	101.1	<.001***	0.871
	TTF $\times$ Visual Cue Presence	1.0	15.0	4.6	0.049*	0.235
	TTS	1.4	21.5	8.5	0.004**	0.363
	Visual Cue Presence	1.0	15.0	7.9	0.013*	0.345
σ_1	TTF	1.0	15.0	95.1	<.001***	0.864
	TTF $\times$ TTS	2.0	29.9	4.9	0.015*	0.246
	TTF $\times$ Visual Cue Presence	1.0	15.0	10.9	0.005**	0.422
	Visual Cue Presence	1.0	15.0	9.0	0.009**	0.376
μ_2	TTS	1.4	21.5	8.5	0.004**	0.363
	Visual Cue Presence	1.0	15.0	7.9	0.013*	0.345
σ_2	TTF	1.0	15.0	10.2	0.006**	0.404
	TTS $\times$ Visual Cue Presence	1.9	28.1	10.0	<.001***	0.400
	Visual Cue Presence	1.0	15.0	10.5	0.005**	0.412

*, **, and *** indicate $p < .05$, $p < .01$, and $p < .001$, respectively.

relatively to the first button input. The fact that σ_2 tended to be lower overall than σ_1 also supports the hypothesis that additional sensory cues exist that contribute to the planning of the second button input (Figure 14d).

Meanwhile, the encoding precision of the internal clock increases as the encoded interval becomes shorter (i.e., Equation 1), and in CTP tasks, TTS is generally significantly shorter than P . In other words, compared to P , TTS is a significantly more reliable interval cue. Furthermore, since we observed that the changing trend of σ_2 was governed by TTS, the cue can even be said to be more reliable than the target movement pattern cue (i.e., Section 3.2.2). However, after TTS becomes sufficiently long and unreliable (after 300 ms in Figure 13b, f), participants appear to begin to rely more on the target movement pattern cue. That is, from then on, as TTS becomes longer, σ_2 decreases, as predicted by Equations 2 and 3.

Similar findings were made in previous studies on human rhythm reproduction performance [34, 37]. Studies have shown that, for example, when we clap to a rhythm of a certain period P , the precision of the clapping can be significantly improved by mentally subdividing the rhythm (e.g., counting internally every $P/2$ or $P/3$)

[35]. Similar to how in CTP users relied more on TTS cue rather than P , this phenomenon occurs because encoding shorter lengths of time chunks allows the internal clock to have higher precision.

6.2 What Factors Determine User Performance in CTP? (RQ2)

One of the most surprising findings from previous temporal pointing research is that button activation point has a significant impact on input timing precision [20, 24, 30]. In particular, previous studies have shown that the keyPressed method enables button input with higher precision than the keyReleased method [20, 24], and the same phenomenon was replicated in this study. In Variations 1 and 2, we confirmed that the second button input showed a significantly lower σ_2 when performed with the keyPressed method (Figure 14). The σ_2 in the keyPressed method was 82.8% (in Variation 1) and 93.7% (in Variation 2) of σ_2 in the keyReleased method, which is consistent with previous studies [20, 24]. Because the first button input for both methods was performed with the KeyPressed, no significant difference was found in σ_1 as expected.

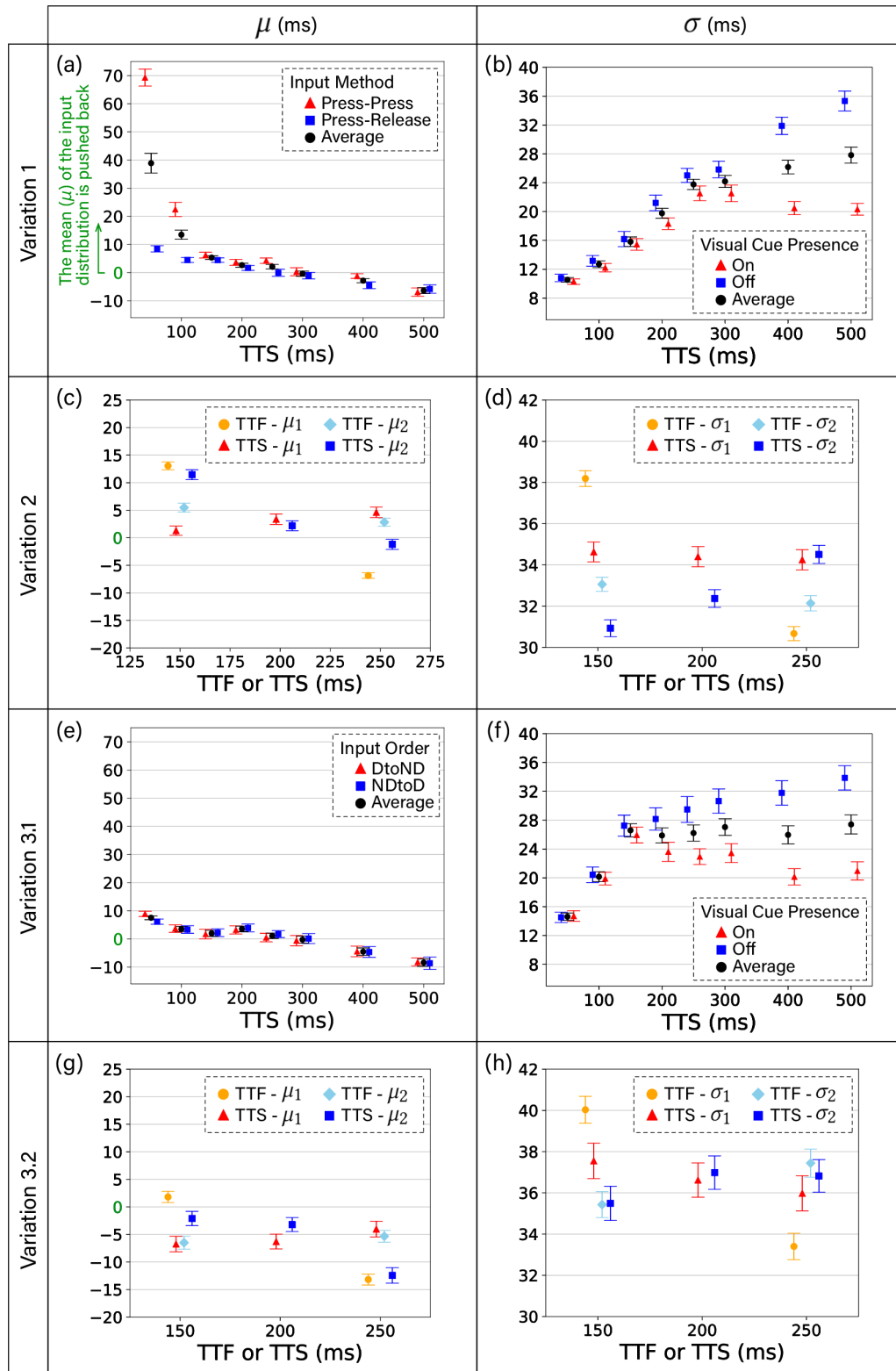


Figure 13: Effects of factors in each study variation - The error bars represent the standard error of the mean.

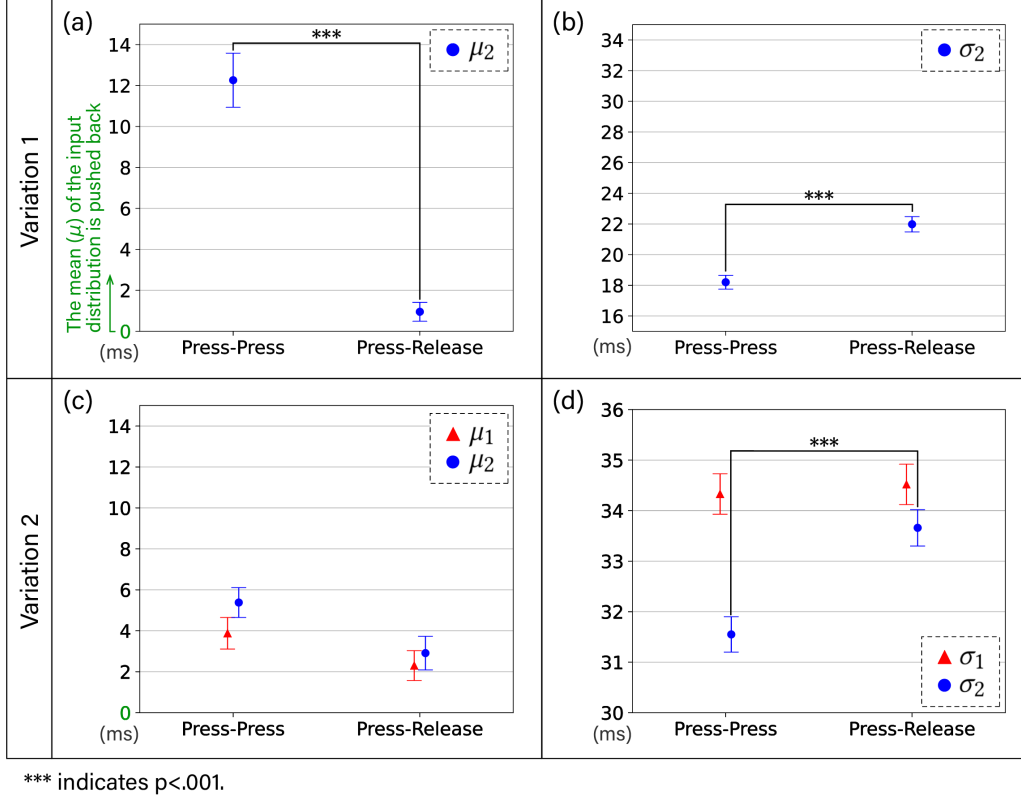


Figure 14: Effects of Input Method in Variations 1 and 2 (Note: Variation 3 does not include Input Method as an independent variable) - The error bars represent the standard error of the mean.

However, it seems that the keyPressed method is not always good in CTP. To activate a button twice, while the keyReleased method requires a single Down-Up movement of the finger, the keyPressed method requires moving the finger in three steps: Down-Up-Down. Due to the time required for this additional motor execution, we observed that in the keyPressed condition, when the interval between button presses is short (e.g., less than 150 ms), the second button press tends to be pushed back significantly on average (Figure 13a). To compensate for the pushback due to motor limitations, the mean of the first button input distribution (μ_1) was pulled forward when TTS was shortened. To minimize this workload, the threshold for double-click activation has typically been set at 100 ms or higher [1]. This phenomenon disappeared in the bimanual condition (Variation 3) where the motor limit was not critical (Figure 13e).

In Variation 3, we also addressed the impact of interlimb coordination (i.e., Input Order) on user performance in the CTP task. As a result, we found that the effects of Input Order on button input distribution were not significant ($p > 0.23$). This contradicts findings from previous studies that the dominant limb or hand is more advantageous in terms of speed and timing consistency of synchronization in rhythmic tasks [6, 32, 38]. We attribute this difference to the fact that the CTP task is not a purely rhythmic

task; in CTP, target movement is visible and the input interval is shorter than that of a typical rhythmic task (e.g., longer than 300 ms). Additionally, the target's movement direction [28], which varied depending on Input Order, may have had a stronger effect on button input performance than handedness.

6.3 Implications and Future Work

This study provides significant implications for interaction design in real-time video games where temporal pointing tasks are frequently given. First, if the game frequently requires consecutive button inputs with intervals of less than 150 ms, the keyReleased method, which can bypass players' motor limitations, should be considered. Otherwise, the keyPressed method should primarily be provided to help players achieve optimal performance. Second, the counterintuitive fact that an increase in TTS in a CTP task leads to a decrease in button input precision must be taken into account in game difficulty design. Considering that CTP is generally used in video games in special situations such as combos or finishing moves, this study opens the way for designers to adjust game difficulty in a more sophisticated, situation-specific manner. In fact, previous studies attempted to adjust only the *overall difficulty* of the game by controlling basic variables of temporal pointing, such as input period, cue-viewing time, and input latency [25, 26].

This study also provides meaningful implications for future modeling studies. Our hypothesis that the interval between two button inputs (i.e., TTS) is utilized as an independent sensory cue can be tested based on the cue-integration theory [11, 21, 23]. The theory provides a general formulation of the process by which multiple sensory cues of different reliability are integrated to result in a single estimate. The moving-target acquisition model [21, 23] used as a baseline in this study was also derived based on the theory.

Meanwhile, unlike existing temporal pointing models, changes in the input distribution mean (μ) must also be seriously considered in CTP modeling. For example, to predict the phenomenon in which the second button input is pushed back due to a short TTS in CTP, we can refer to the Wing-Kristofferson model [40, 41], which describes the fundamental delay of the motor response triggered by the internal clock. We also expect future models of μ to take into account the success reward each button press gives the user; users will want to perform button inputs with higher rewards more accurately (with lower errors). That is, μ_1 and μ_2 in CTP will be modulated by the difference in reward between button inputs. There was a previous study that suggested such a reward maximization mechanism as the reason why μ is generally observed to be negative in ordinary temporal pointing (i.e., negative mean asynchrony) [36]. Another study [2] that built an economic model to predict pointing error rates may also provide useful insights.

6.4 Limitations

While we investigated user performance in CTP through three study variations, there are other variations we have not covered. For example, we can imagine scenarios where users become aware of the activation timing of the second button input after they perform the first button input. Additionally, there might be situations where Visual Cue Presence is set to On only in either the time section up to the first button input or in the section between the first and second button inputs. In our experiment, the target (a moving line) moved in one dimension with a constant speed, but real-world applications often involve more complex and unpredictable target motions. We have limitations in recruiting participants with diverse demographics. Although the crucial effect of age on performance was reported in ordinary temporal pointing [21], our participants were mostly young, and in Variation 3, most were right-handed. We did not measure the ordinary temporal pointing performance of each participant as a baseline.

7 CONCLUSION

The CTP task requires users to generate two consecutive button inputs at each timing specified by the system with a short time interval, considering them as a single chunk. Despite the extensive use of CTP in many HCI scenarios, user performance in CTP has barely been touched upon, and there is even a lack of understanding of the task itself. To lay the foundation for the in-depth understanding of user performance in CTP, we first established the task taxonomy that classifies its task scenarios into two types. We then investigated user performance in CTP via the user study involving a total of 100 participants. CTP may be considered as the concatenation of two ordinary temporal pointing processes. However, through our analysis, we concluded that CTP is a unique

task that cannot be considered as two ordinary temporal pointing processes, exhibiting a contrasting trend in the button input distribution. We also identified diverse factors that contribute to the performance, including input method, motor limitations, and different hand use.

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